Lab 16 — Analyse Statistics, Correlation and Rolling Risk

Python Programming for Finance · TGS-2026065049 · LO7

Scenario

A portfolio risk analyst must profile returns, compare bank-stock relationships and track changing volatility over time.

Goal

A statistics table, correlation matrix, rolling-volatility series and drawdown chart.

Tools

Pandas, NumPy, Matplotlib, CSV

Step-by-step

1. Load multi_asset_prices.csv and calculate daily returns.

2. Compute count, mean, median, standard deviation and quantiles.

3. Calculate covariance and correlation matrices.

```
python starter.py
```

4. Calculate 20-day rolling volatility and annualise using sqrt(252).

5. Calculate cumulative return and drawdown.

```
python solution.py
```

6. Check correlation symmetry and drawdown sign conventions.

```
python -m unittest test_solution.py
```

Acceptance criteria

Correlation is symmetric with 1 on the diagonal, rolling volatility has the expected warm-up NaNs, and maximum drawdown is non-positive.

Evidence

Capture the executed output, validation result, generated file or chart, and a one-sentence finance interpretation.

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